Chapter 07 Part 1: Solutions

1.

Solution
$0.1\ M\ HC_2H_3O_2(aq)$
$0.1\ M\ KI(aq)$
$0.1\ M\ CH_3OH(aq)$

Of the three solutions listed in the table above, which one, if any, has the greatest electrical conductivity and why?

(A) $0.1\ M\ HC_2H_3O_2(aq)$ because its molecules have the most atoms.

(B) $0.1\ M\ KI(aq)$ because KI completely dissociates in water to produce ions.

(C) $0.1\ M\ CH_3OH(aq)$ because its molecules can form hydrogen bonds.

(D) All three solutions have the same electrical conductivity because the concentrations are the same.

2. Solid $Al(NO_3)_3$ is added to distilled water to produce a solution in which the concentration of nitrate, $[NO_3^-]$, is 0.10 *M*. What is the concentration of aluminum ion, $[Al^{3+}]$, in this solution?

(A) 0.010 *M*

(B) 0.033 *M*

(C) 0.066 *M*

(D) 0.10 *M*

(E) 0.30 *M*

3. Which of the following questions about the components in a mixture could be investigated with a paper chromatography experiment?

(A) Do the components have different densities?

(B) Do the components have different molecular masses?

(C) Do the components have different molecular polarities?

(D) Do the components have different average molecular speeds?

4. Which of the following is the best piece of laboratory glassware for preparing 500.0 mL of an aqueous solution of a solid?

(A) Volumetric flask

(B) Erlenmeyer flask

(C) Test tube

(D) Graduated beaker

(E) Graduated cylinder

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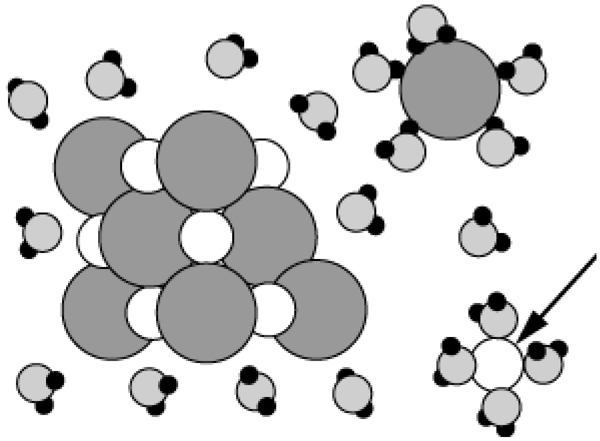
5.



An ionic crystal is added to water and starts to dissolve as shown in the particulate representation above. Based on the orientation of the water molecules, what can be assumed about the charge of the hydrated particle indicated by the arrow?

(A) The particle is positively charged, because the oxygen atoms in water are attracted to it.

(B) The particle is negatively charged, because hydrogen atoms in water are attracted to it.

(C) The particle is both positively and negatively charged, because the water molecules are attracted to it.

(D) The particle is neutral, because the dipoles of the water molecules neutralize the charge of the particle.

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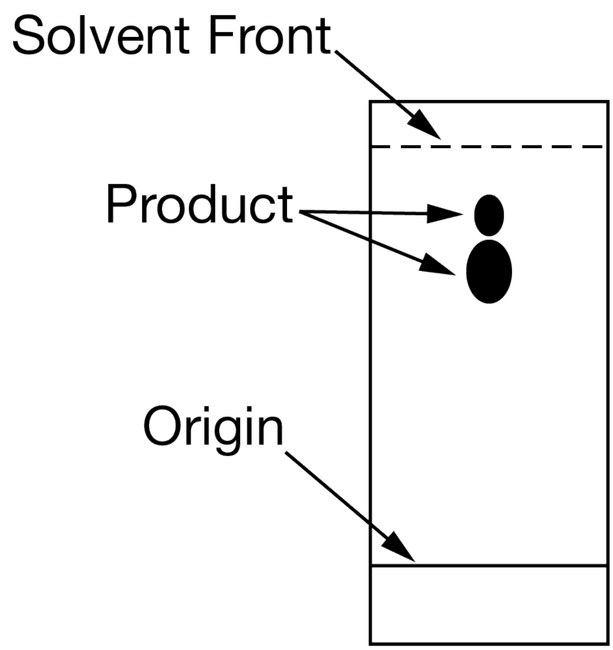
6.



Solvent	Boiling Point (°C)	Relative Polarity
Pentane	36	0.1
Hexane	68	0.1
Ethyl acetate	77	4.4
Methanol	65	5.1

The diagram above shows a thin-layer chromatogram of a mixture of products from a chemical reaction. The separation was performed using 50% ethyl acetate in hexane as the solvent (mobile phase) and silica gel as the polar stationary phase. On the basis of the chromatogram and the information about solvents in the table above, which of the following would be the best way to decrease the distance that the products travel up the plate?

(A) Use pentane instead of hexane in the solvent.

(B) Decrease the percentage of ethyl acetate in the solvent.

(C) Increase the percentage of ethyl acetate in the solvent.

(D) Add up to 5% methanol to the solvent.

7. A 0.20 mol sample of $MgCl_2(s)$ and a 0.10 mol sample of $KCl(s)$ are dissolved in water and diluted to 500 mL. What is the concentration of Cl^- in the solution?

(A) 0.15 *M*

(B) 0.30 *M*

(C) 0.50 *M*

(D) 0.60 *M*

(E) 1.0 *M*

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A student uses visible spectrophotometry to determine the concentration of $CoCl_2(aq)$ in a sample solution. First the student prepares a set of $CoCl_2(aq)$ solutions of known concentration. Then the student uses a spectrophotometer to determine the absorbance of each of the standard solutions at a wavelength of $510\ nm$ and constructs a standard curve. Finally, the student determines the absorbance of the sample of unknown concentration.

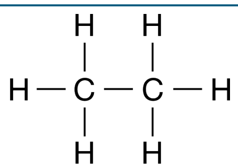
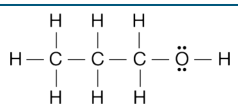
8. The original solution used to make the solutions for the standard curve was prepared by dissolving $2.60\ g$ of $CoCl_2$ (molar mass $130.\ g/mol$) in enough water to make $100.\ mL$ of solution. What is the molar concentration of the solution?

(A) $0.200\ M$

(B) $0.500\ M$

(C) $1.00\ M$

(D) $5.00\ M$

Compound	**Molar Mass**	**Balanced Equation for the Combustion of One Mole of the Compound**	ΔH°_{comb} (kJ/mol_{rxn})
 Ethane	$30.\ g/mol$	$C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$	-1560
 Propanol	$60.\ g/mol$	$C_3H_8O(l) + \frac{9}{2}O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$	-2020
Unknown	?	$? +\ 5\,O_2(g) \rightarrow 4CO_2(g) + 4H_2O(l)$	-2240

9. Based on the structural formulas, which of the following identifies the compound that is more soluble in water and best helps to explain why?

(A) Ethane, because the electron clouds of its molecules are more polarizable than those of propanol.

(B) Ethane, because its molecules can fit into the spaces between water molecules more easily than those of propanol can.

(C) Propanol, because its molecules have a greater mass than the molecules of ethane have.

(D) Propanol, because its molecules can form hydrogen bonds with water molecules but those of ethane cannot.

10. If 50. mL of 1.0 M NaOH is diluted with distilled water to a volume of 2.0 L, the concentration of the resulting solution is

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(A) 0.025 M
(B) 0.050 M
(C) 0.10 M
(D) 0.50 M
(E) 1.0 M

11. If 200. mL of 0.60 M $MgCl_2(aq)$ is added to 400. mL of distilled water, what is the concentration of $Mg^{2+}(aq)$ in the resulting solution? (Assume volumes are additive).

(A) 0.20 M
(B) 0.30 M
(C) 0.40 M
(D) 0.60 M
(E) 1.2 M

12. What is the molarity of $I^-(aq)$ in a solution that contains 34 g of SrI_2 (molar mass 341 g) in 1.0 L of the solution?

(A) 0.034 M
(B) 0.068 M
(C) 0.10 M
(D) 0.20 M
(E) 0.68 M

13. When 70. milliliter of 3.0-molar Na_2CO_3 is added to 30. milliliters of 1.0-molar $NaHCO_3$ the resulting concentration of Na^+ is

(A) 2.0 M
(B) 2.4 M
(C) 4.0 M
(D) 4.5 M
(E) 7.0 M

14. A 40.0 mL sample of 0.25 M KOH is added to 60.0 mL of 0.15 M $Ba(OH)_2$. What is the molar concentration of $OH^-(aq)$ in the resulting solution? (Assume that the volumes are additive.)

(A) 0.10 M
(B) 0.19 M
(C) 0.28 M
(D) 0.40 M
(E) 0.55 M

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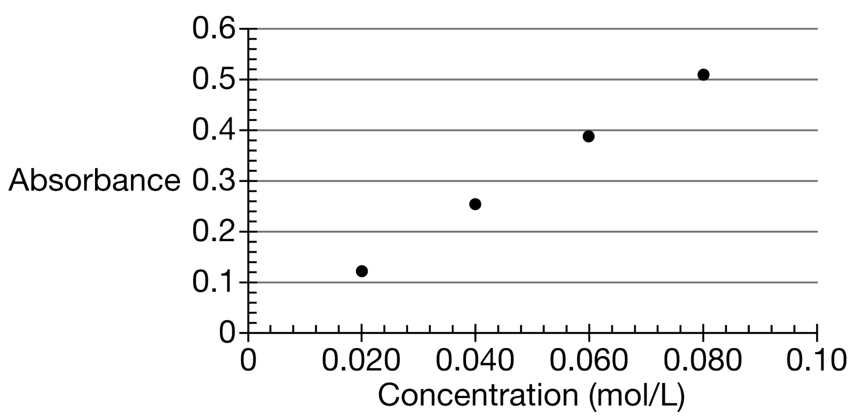
An experiment was performed to investigate the reaction between Zn metal and $Ni^{2+}(aq)$ at different concentrations. Because the $Ni^{2+}(aq)$ ion is green, the extent of the reaction was determined using spectrophotometric analysis. Four $20.0\ mL$ standard solutions of $Ni^{2+}(aq)$ were prepared by dissolving $NiCl_2 \cdot 6H_2O$ (molar mass $240\ g/mol$) in water. The absorbance of each solution was measured; the results are shown both in the table below and in the following plot of the absorbance data.

Solution	$[Ni^{2+}]$	Absorbance
1	0.020	0.12
2	0.040	0.25
3	0.060	0.39
4	0.080	0.51



15. The concentration of $Cl^-(aq)$ ions in solution 2 was

(A) $0.020\ M$

(B) $0.040\ M$

(C) $0.060\ M$

(D) $0.080\ M$

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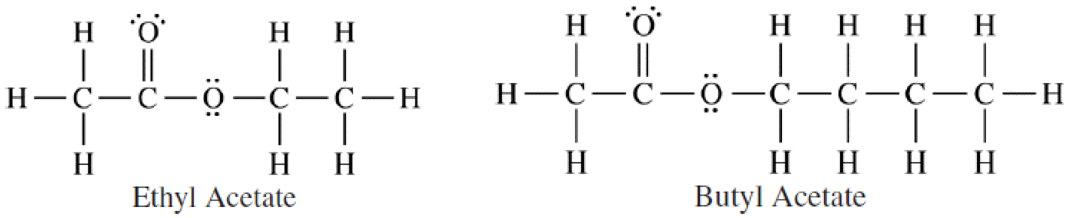
16.



A mixture containing equal numbers of moles of ethyl acetate and butyl acetate was separated using distillation. Based on the diagrams shown above, which of the following identifies the substance that would be initially present in higher concentration in the distillate and correctly explains why that occurs?

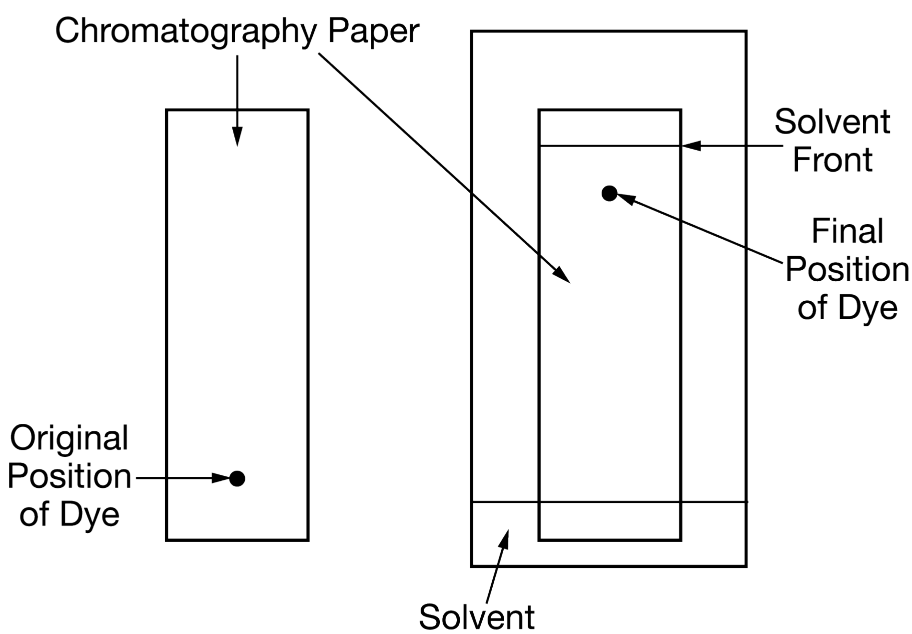
(A) Ethyl acetate, because it has fewer C-C bonds to break

(B) Ethyl acetate, because it has a shorter carbon chain and weaker London dispersion forces

(C) Butyl acetate, because it has more C-C bonds to break

(D) Butyl acetate, because it has a longer carbon chain and weaker dipole-dipole attractions

17.



Based on the results of the paper chromatography experiment shown above, which of the following can be concluded about the dye?

(A) It has a small molar mass.

(B) It has weak intermolecular forces.

(C) It has a weaker attraction for the stationary phase that it has for the mobile phase.

(D) It has a stronger attraction for the stationary phase than it has for the mobile phase.

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18. A student obtains a liquid sample of green food coloring that is known to contain a mixture of two solid pigments, one blue and one yellow, dissolved in an aqueous solution of ethanol. Which of the following laboratory setups is most appropriate for the student to use in order to separate and collect a substantial sample of each of the two pigments?

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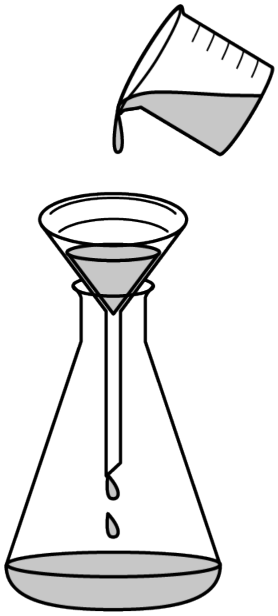
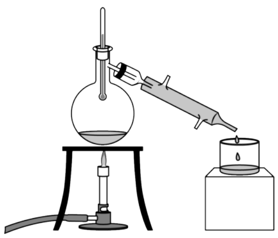
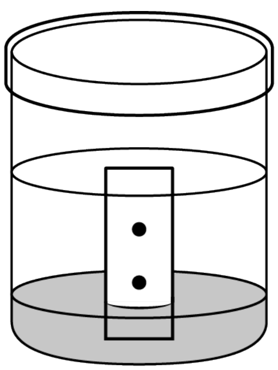
(A)



(B)



(C)



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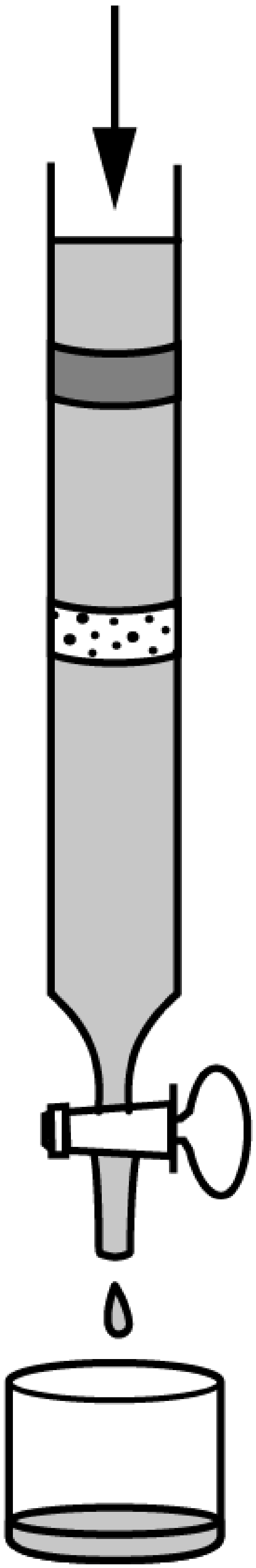
(D)



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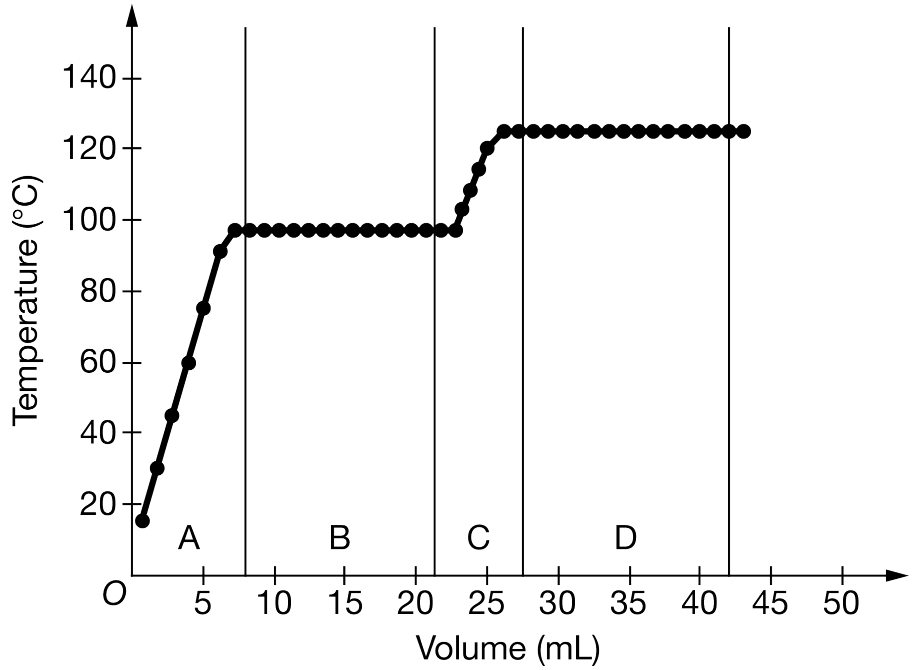
19.



A student performed a fractional distillation of a mixture of two straight-chain hydrocarbons, C_7H_{16} and C_8H_{18}. Using four clean, dry flasks, the student collected the distillate over the volume ranges (A, B, C, and D) shown in the graph above. Over what volume range should the student collect the distillate of the compound with the stronger intermolecular forces?

(A) A
(B) B
(C) C
(D) D

20. On the basis of molecular structure and bond polarity, which of the following compounds is most likely to have the greatest solubility in water?

(A) CH_4
(B) CCl_4
(C) NH_3
(D) PH_3

21. Which of the following pieces of laboratory glassware should be used to most accurately measure out a 25.00 mL sample of a solution?

(A) 5 mL pipet
(B) 25 mL pipet
(C) 25 mL beaker
(D) 25 mL Erlenmeyer flask
(E) 50 mL graduated cylinder

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22. A student wishes to prepare 2.00 liters of 0.100–molar KIO_3 (molecular weight 214). The proper procedure is to weigh out

(A) 42.8 grams of KIO_3 and add 2.00 kilograms of H_2O

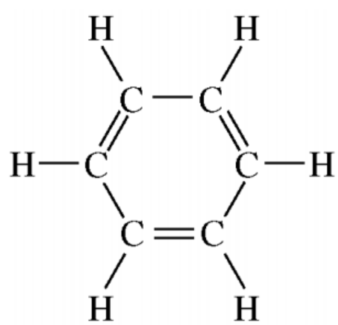
(B) 42.8 grams of KIO_3 and add H_2O until the final homogeneous solution has a volume of 2.00 liters

(C) 21.4 grams of KIO_3 and add H_2O until the final homogeneous solution has a volume of 2.00 liters

(D) 42.8 grams of KIO_3 and add 2.00 liters of H_2O

(E) 21.4 grams of KIO_3 and add 2.00 liters of H_2O

23.



Benzene, C_6H_6 , has the structure shown above. Considering the observation that benzene is only sparingly soluble in water, which of the following best describes the intermolecular forces of attraction between water and benzene?

(A) Benzene is nonpolar, therefore there are no forces between water and benzene.

(B) The H atoms in benzene form hydrogen bonds with the O atoms in water.

(C) Benzene is hydrophobic, therefore there is a net repulsion between water and benzene.

(D) There are dipole-induced dipole and London dispersion interactions between water and benzene.

24. For an experiment, a student needs 100.0 mL of 0.4220 M NaCl. If the student starts with NaCl(s) and distilled water, which of the following pieces of laboratory glassware should the student use to prepare the solution with the greatest accuracy?

(A) 25 mL volumetric pipet

(B) 100 mL Erlenmeyer flask

(C) 100 mL graduated cylinder

(D) 100 mL volumetric flask

(E) 1 L beaker

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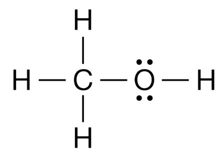
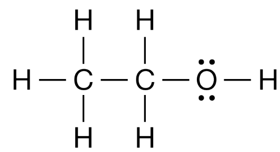
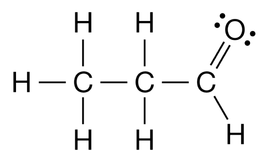
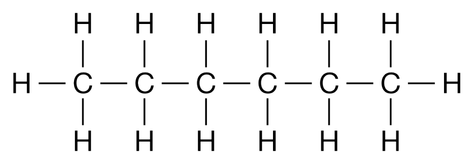
Compound	Structure	Boiling Point (°C)
Methanol		65
Ethanol		78
Propanal		49
Hexane		69

25. A $50\ \mathrm{mL}$ sample of $C_6H_{14}(l)$ is mixed with a $50\ \mathrm{mL}$ sample of $H_2O(l)$, and the mixture is shaken vigorously. The two liquids do not stay mixed but instead form two separate layers. The density of hexane is $0.66\ \mathrm{g/mL}$, and the density of water is $1.00\ \mathrm{g/mL}$. A $1.0\ \mathrm{g}$ sample of $I_2(s)$ is added to the mixture, which is shaken again. Which of the following best predicts what happens to the $I_2(s)$?

(A) I_2 will be found mainly in the top layer because it will dissolve more in the $H_2O(l)$.

(B) I_2 will be found mainly in the bottom layer because it will dissolve more in the $H_2O(l)$.

(C) I_2 will be found mainly in the top layer because it will dissolve more in the $C_6H_{14}(l)$.

(D) I_2 will be found mainly in the bottom layer because it will dissolve more in the $C_6H_{14}(l)$.

26. Of the following organic compounds, which is LEAST soluble in water at 298 K?

(A) CH_3OH, methanol

(B) $CH_3CH_2CH_2OH$, l-propanol

(C) C_6H_{14}, hexane

(D) $C_6H_{12}O_6$, glucose

(E) CH_3COOH, ethanoic (acetic) acid

27. Which of the following molecules is least soluble in water?

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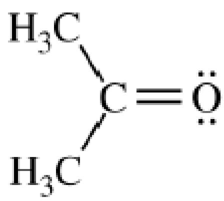
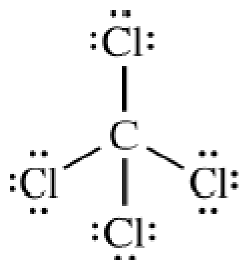
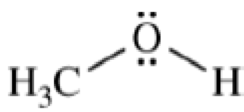
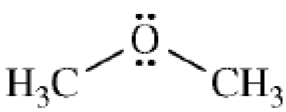
(A) 

(B) 

(C) 

(D) 

28. Approximately what mass of $CuSO_4 \cdot 5H_2O$ (250 g mol^{-1}) is required to prepare 250 mL of 0.10 *M* copper(II) sulfate solution?

(A) 4.0 g
(B) 6.2 g
(C) 34 g
(D) 85 g
(E) 140 g

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The following questions refer to the below.

M^+ is an unknown metal cation with a +1 charge. A student dissolves the chloride of the unknown metal, MCl, in enough water to make 100.0 mL of solution. The student then mixes the solution with excess $AgNO_3$ solution, causing AgCl to precipitate. The student collects the precipitate by filtration, dries it, and records the data shown below. (The molar mass of AgCl is 143 g/mol.)

Mass of unknown chloride, MCl	0.74 g
Mass of filter paper	0.80 g
Mass of filter paper plus AgCl precipitate	2.23 g

29. Which of the following diagrams best represents the $AgNO_3$ solution before the reaction occurs?

Note: water molecules are represented by the symbol

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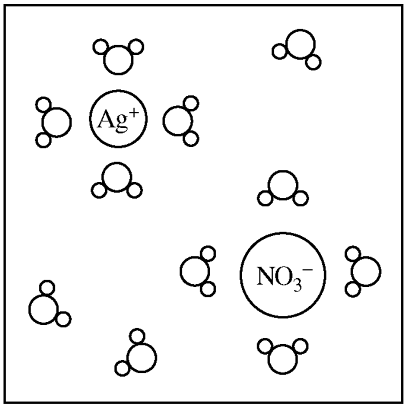
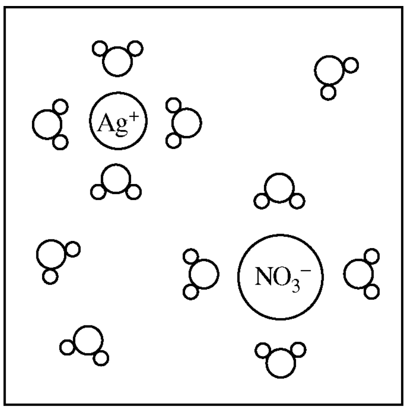
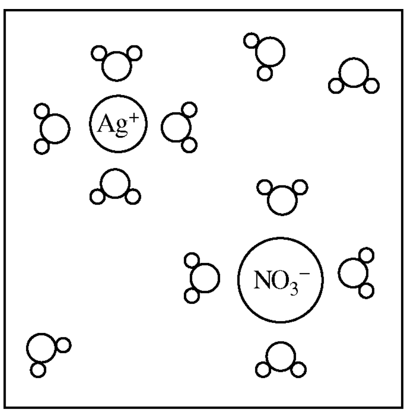
(A)



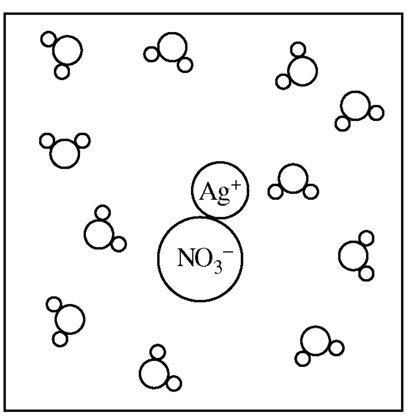
(B)



(C)



(D)



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30. A 360. mg sample of aspirin, $C_9H_8O_4$, (molar mass 180. g), is dissolved in enough water to produce 200. mL of solution. What is the molarity of aspirin in a 50. mL sample of this solution?

(A) 0.0800 *M*
(B) 0.0400 *M*
(C) 0.0200 *M*
(D) 0.0100 *M*
(E) 0.00250 *M*

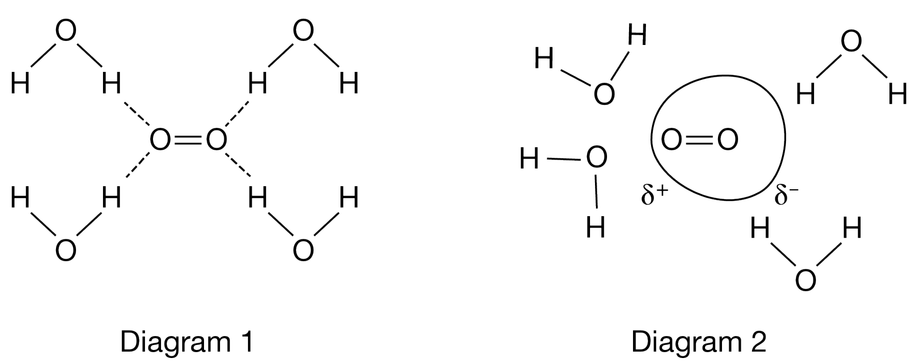
31. A student prepares a solution by dissolving 60.00 g of glucose (molar mass 180.2 g mol^{-1}) in enough distilled water to make 250.0 mL of solution. The molarity of the solution should be reported as

(A) 12.01 *M*
(B) 12.0 *M*
(C) 1.332 *M*
(D) 1.33 *M*
(E) 1.3 *M*

32. How many moles of Na^+ ions are in $100.\ \mathrm{mL}$ of $0.100\ M\ \mathrm{Na_3PO_4}(aq)$?

(A) $0.300\ \mathrm{mol}$
(B) $0.100\ \mathrm{mol}$
(C) $0.0300\ \mathrm{mol}$
(D) $0.0100\ \mathrm{mol}$

33.



The survival of aquatic organisms depends on the small amount of $\mathrm{O_2}$ that dissolves in $\mathrm{H_2O}$. The diagrams above represent possible models to explain this phenomenon. Which diagram provides the better particle representation for the solubility of $\mathrm{O_2}$ in $\mathrm{H_2O}$, and why?

(A) Diagram 1, because $\mathrm{O_2}$ molecules can form hydrogen bonds with the $\mathrm{H_2O}$ molecules.
(B) Diagram 1, because $\mathrm{O_2}$ and $\mathrm{H_2O}$ are polar molecules that can interact through dipole-dipole forces.
(C) Diagram 2, because the polar $\mathrm{H_2O}$ molecules can induce temporary dipoles on the electron clouds of $\mathrm{O_2}$ molecules.
(D) Diagram 2, because the nonpolar $\mathrm{O_2}$ molecules can induce temporary dipoles on the electron clouds of $\mathrm{H_2O}$ molecules.

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Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits each statement. A choice may be used once, more than once, or not at all in each set.

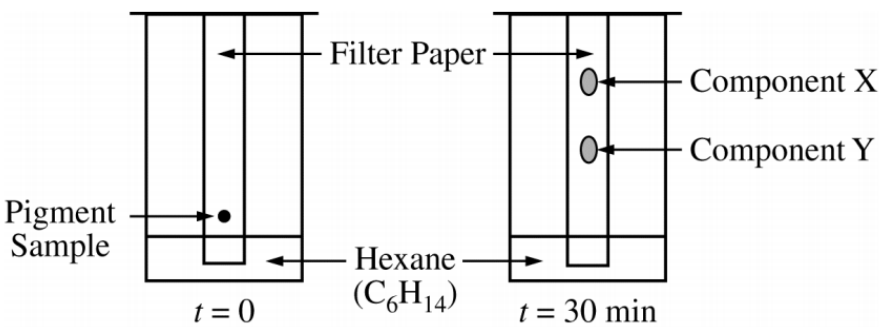
Refer to the following compounds.

(A) $CH_3CH_2CH_2CH_3$
(B) $CH_3CH_2CH_2OH$
(C) CH_3COCH_3
(D) CH_3COOH
(E) $CH_3CH_2CH_2NH_2$

34. Is the LEAST soluble in water

(A) $CH_3CH_2CH_2CH_3$
(B) $CH_3CH_2CH_2OH$
(C) CH_3COCH_3
(D) CH_3COOH
(E) $CH_3CH_2CH_2NH_2$

35.



In a paper chromatography experiment, a sample of a pigment is separated into two components, X and Y, as shown in the figure above. The surface of the paper is moderately polar. What can be concluded about X and Y based on the experimental results?

(A) X has a larger molar mass than Y does.
(B) Y has a larger molar mass than X does.
(C) X is more polar than Y.
(D) Y is more polar than X.

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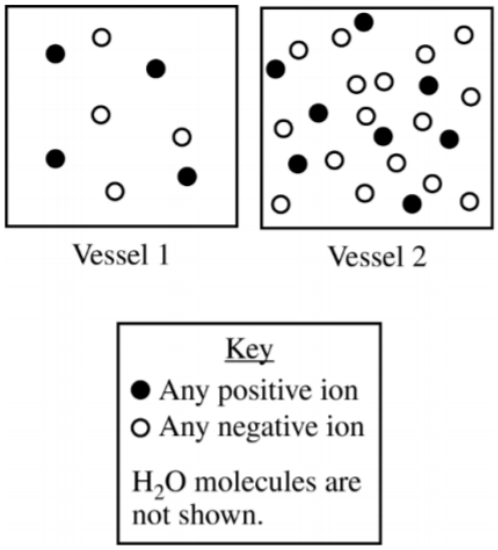
36.



Equal volumes of solutions in two different vessels are represented above. If the solution represented in vessel 1 is $KCl(aq)$, then the solution represented in vessel 2 could be an aqueous solution of

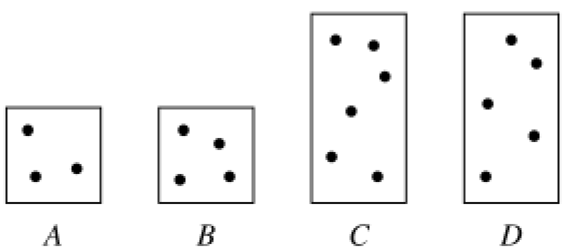
(A) KCl with the same molarity as the solution in vessel 1

(B) KCl with twice the molarity of the solution in vessel 1

(C) $CaCl_2$ with the same molarity as the solution in vessel 1

(D) $CaCl_2$ with twice the molarity of the solution in vessel 1

37.



The diagrams above represent aqueous solutions of compound X at various molarities. Each dot represents one mole of the solute, and the volume of solution in the large boxes is twice the volume in the small boxes. Which solution has the highest molarity?

(A) A

(B) B

(C) C

(D) D

38. How many grams of $\mathrm{CaCl_2}$ (molar mass $= 111\ \mathrm{g/mol}$) are needed to prepare $100.\ \mathrm{mL}$ of $0.100\ M\ \mathrm{Cl^-}(aq)$ ions?

(A) 0.555 g
(B) 1.11 g
(C) 2.22 g
(D) 5.55 g

39.

CH_3—CH_2—CH_2—CH_2—$CH_3$$CH_3$—$CH_2$—$CH_2$—$CH_2$—OHHO—$CH_2$—$CH_2$—$CH_2$—OH

X Y Z

Based on concepts of polarity and hydrogen bonding, which of the following sequences correctly lists the compounds above in the order of their increasing solubility in water?

(A) $Z < Y < X$
(B) $Y < Z < X$
(C) $Y < X < Z$
(D) $X < Z < Y$
(E) $X < Y < Z$

40.

Compound	**Formula**	**Boiling Point** $(^\circ C)$	**Density** (g/mL)
Hexane	C_6H_{14}	69	0.66
Octane	C_8H_{18}	126	0.70

A student obtains a mixture of the liquids hexane and octane, which are miscible in all proportions. Which of the following techniques would be best for separating the two components of the mixture, and why?

(A) Filtration, because the different densities of the liquids would allow one to pass through the filter paper while the other would not.
(B) Paper chromatography, because the liquids would move along the stationary phase at different rates owing to the difference in polarity of their molecules.
(C) Column chromatography, because the higher molar mass of octane would cause it to move down the column faster than hexane.
(D) Distillation, because the liquids would boil at different temperatures owing to the difference in strength of their intermolecular forces.

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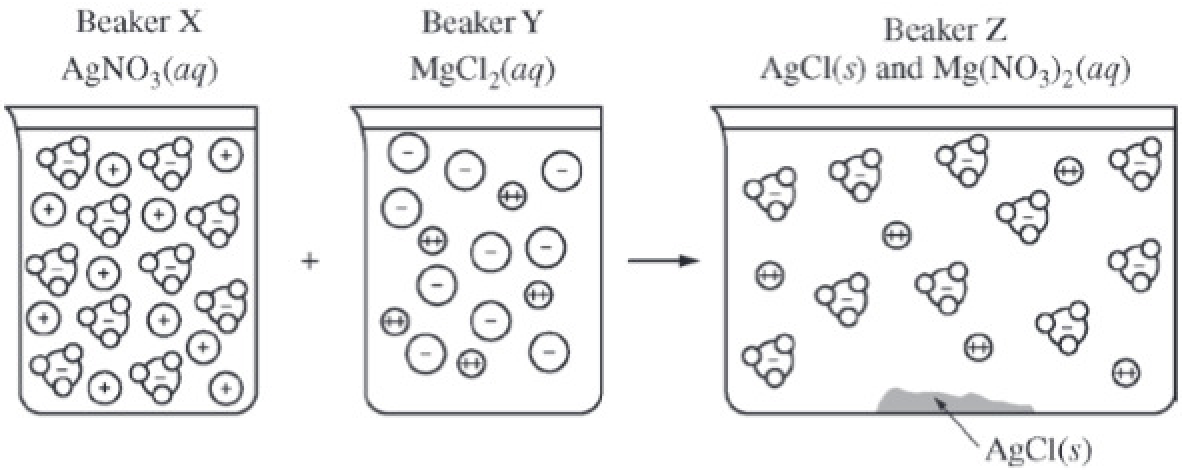
41.



Beaker X and beaker Y each contain 1.0 L of solution, as shown above. A student combines the solutions by pouring them into a larger, previously empty beaker Z and observes the formation of a white precipitate. Assuming that volumes are additive, which of the following sets of solutions could be represented by the diagram above?

Beaker X | Beaker Y| Beaker Z

(A) 2.0 M $AgNO_3$ | 2.0 M $MgCl_2$ | 4.0 M $Mg(NO_3)_2$ and AgCl(s)

(B) 2.0 M $AgNO_3$ | 2.0 M $MgCl_2$ | 2.0 M $Mg(NO_3)_2$ and AgCl(s)

(C) 2.0 M $AgNO_3$ | 1.0 M $MgCl_2$ | 1.0 M $Mg(NO_3)_2$ and AgCl(s)

(D) 2.0 M $AgNO_3$ | 1.0 M $MgCl_2$ | 0.50 M $Mg(NO_3)_2$ and AgCl(s)

42. Sodium chloride is LEAST soluble in which of the following liquids?

(A) H_2O

(B) CCl_4

(C) HF

(D) CH_3OH

(E) CH_3COOH

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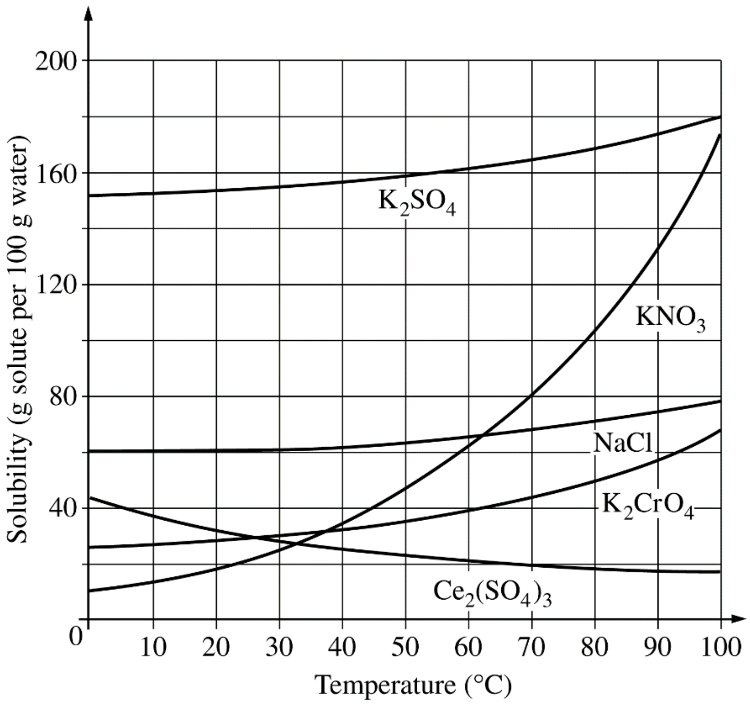
43.



On the basis of the solubility curves shown above, the greatest percentage of which compound can be recovered by cooling a saturated solution of that compound from 90°C to 30°C?

(A) NaCl

(B) KNO_3

(C) K_2CrO_4

(D) K_2SO_4

(E) $Ce_2(SO_4)_3$

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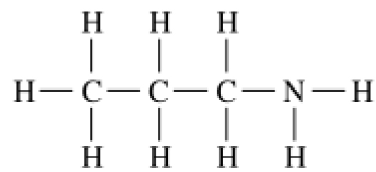
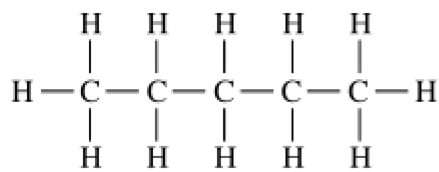
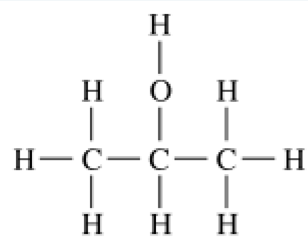
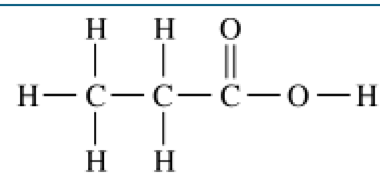
Substance	Structural Formula	Molar Mass (g/ mol)
Propylamine		59
Pentane		72
2-propanol		60
Propanoic acid		74

A student was given samples of four different unknown liquids. The unknown liquids are propylamine, pentane, 2-propanol, and propanoic acid. The structures and molar masses of the compounds are shown in the table above. On the basis of this information, the student designed some experiments to identify the samples.

44. The student mixed samples of each of the four compounds with water. Only one of the four did not dissolve in water. Based on the structures of the molecules and the strengths and types of their intermolecular forces, the compound that did not dissolve is most likely

(A) propylamine

(B) pentane

(C) 2-propanol

(D) propanoic acid

45. Based on their Lewis diagrams, which of the following pairs of liquids are most soluble in each other?

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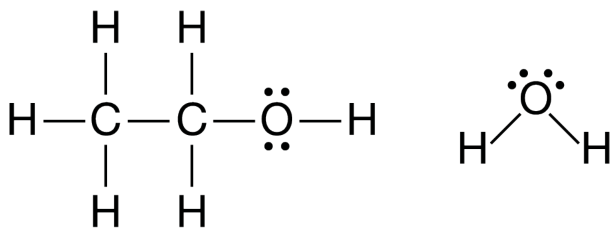
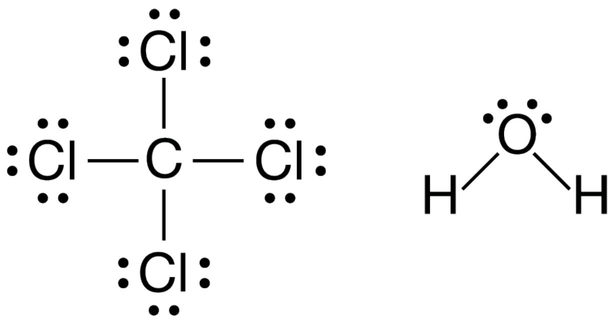
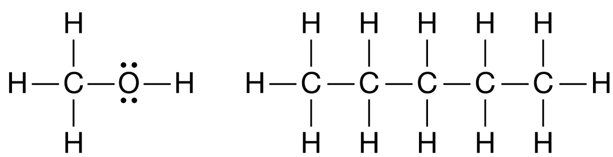
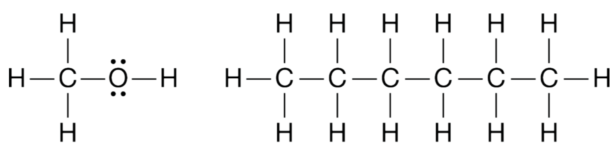
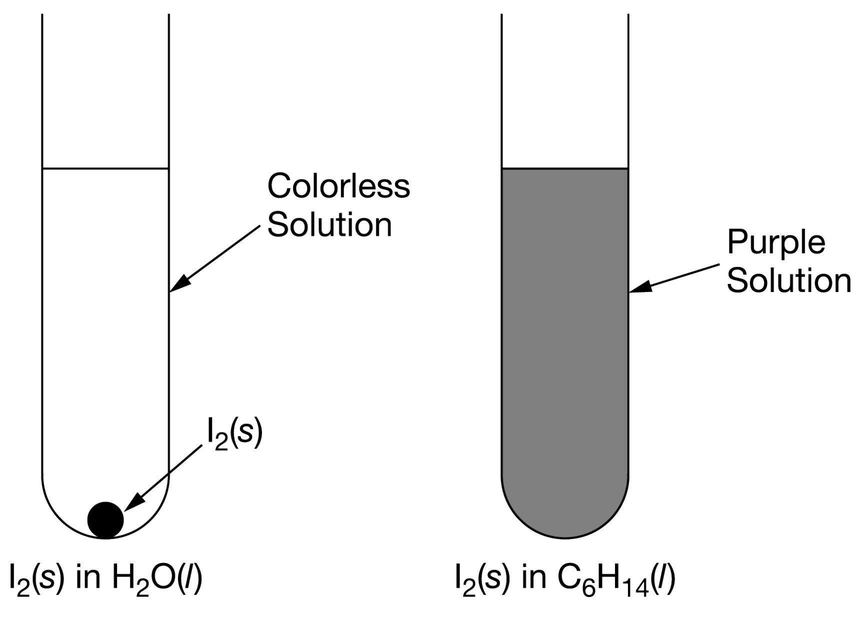
(A) 

(B) 

(C) 

(D) 

46.



A student places a piece of $I_2(s)$ in $50.0\ \ mL$ of $H_2O(l)$, another piece of $I_2(s)$ of the same mass in $50.0\ \ mL$ of $C_6H_{14}(l)$, and shakes the mixtures. The results are shown above. What do the results indicate about the intermolecular interactions of the substances?

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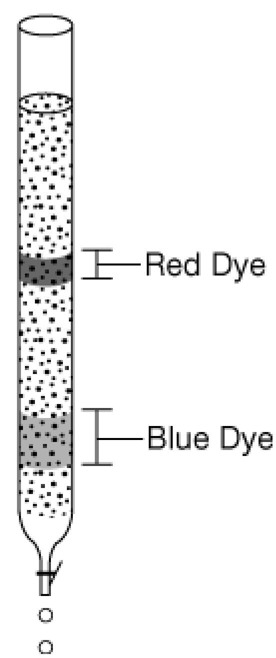
(A) I_2 and H_2O have similar intermolecular interactions, and I_2 and C_6H_{14} do not.
(B) I_2 and C_6H_{14} have similar intermolecular interactions, and I_2 and H_2O do not.
(C) I_2, H_2O, and C_6H_{14} all have similar intermolecular interactions.
(D) I_2, H_2O, and C_6H_{14} have three completely different types of intermolecular interactions.

47. Which of the following substances has the greatest solubility in $C_5H_{12}(l)$ at 1 atm?

(A) $SiO_2(s)$
(B) $NaCl(s)$
(C) $H_2O(l)$
(D) $CCl_4(l)$
(E) $NH_3(g)$

48. Which of the following techniques is most appropriate for the recovery of solid KNO_3 from an aqueous solution of KNO_3?

(A) Paper chromatography
(B) Filtration
(C) Titration
(D) Electrolysis
(E) Evaporation to dryness



A student placed a sample of a food coloring that contains a mixture of a blue dye and a red dye at the top of a chromatography column filled with a nonpolar stationary phase. When water is poured through the column, two bands of colors are seen in the column, as shown in the diagram above.

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49. Which of the following will most likely be observed if $CH_3OH(l)$ is used in the column instead of water? (Assume that both solvents flow through the column at the same rate.)

(A) The dyes will not pass through the column because $CH_3OH(l)$ is a nonpolar solvent.

(B) The dyes will pass through the column at a slower rate because $CH_3OH(l)$ is a less polar solvent than water.

(C) The dyes will pass through the column at a faster rate because $CH_3OH(l)$ has a greater molar mass than water.

(D) No differences will be observed because the polarity of the solvent does not affect the separation of the dyes.

---

50. The volume of distilled water that should be <u>added</u> to 10.0 mL of 6.00 *M* HCl(*aq*) in order to prepare a 0.500 *M* HCl(*aq*) solution is approximately

(A) 50.0 mL

(B) 60.0 mL

(C) 100. ML

(D) 110. ML

(E) 120. mL

51. The volume of water that must be added in order to dilute 40 mL of 9.0 *M* HCl to a concentration of 6.0 *M* is closest to

(A) 10 mL

(B) 20 mL

(C) 30 mL

(D) 40 mL

(E) 60 mL

52. How many mL of 10.0 *M* HCl are needed to prepare 500. mL of 2.00 *M* HCl ?

(A) 1.00 mL

(B) 10.0 mL

(C) 20.0 mL

(D) 100. mL

(E) 200. mL

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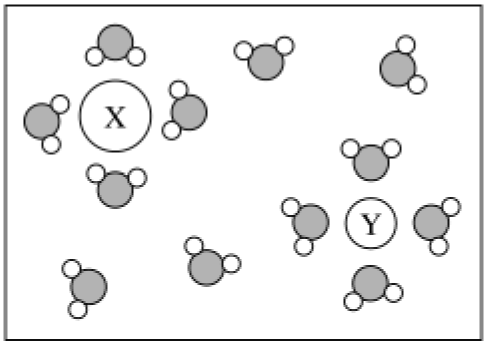
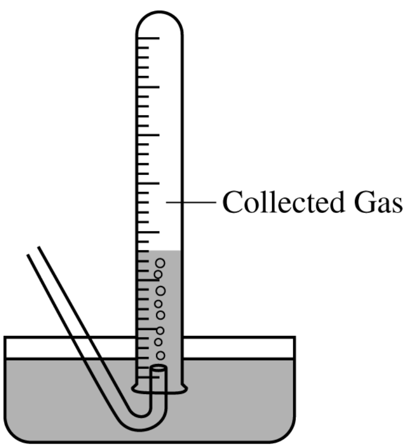
53.



The diagram above represents a solution of ions in water. Which of the following are the most likely identities of ions X and Y ?

(A) Ion X is Na^+, and ion Y is K^+.
(B) Ion X is Na^+, and ion Y is F^-.
(C) Ion X is Cl^-, and ion Y is K^+.
(D) Ion X is Cl^-, and ion Y is F^-.

54.



Gases generated in a chemical reaction are sometimes collected by the displacement of water, as shown above. Which of the following gases can be quantitatively collected by this method?

(A) H_2
(B) CO_2
(C) HCl
(D) SO_2
(E) NH_3

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55. A 0.10 *M* aqueous solution of sodium sulfate, Na_2SO_4 , is a better conductor of electricity than a 0.10 *M* aqueous solution of sodium chloride, NaCl. Which of the following best explains this observation?

(A) Na_2SO_4 is more soluble in water than NaCl is.

(B) Na_2SO_4 has a higher molar mass than NaCl has.

(C) To prepare a given volume of 0.10 *M* solution, the mass of Na_2SO_4 needed is more than twice the mass of NaCl needed.

(D) More moles of ions are present in a given volume of 0.10 *M* Na_2SO_4 than in the same volume of 0.10 *M* NaCl.

(E) The degree of dissociation of Na_2SO_4 in solution is significantly greater than that of NaCl.

56. Which of the following substances is a strong electrolyte when dissolved in water?

(A) Sucrose

(B) Ethanol

(C) Sodium nitrate

(D) Acetic acid

(E) Ammonia

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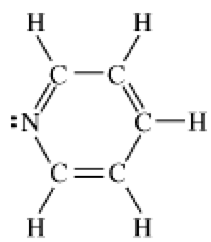
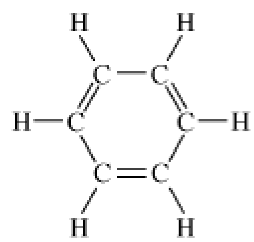
57. Answer the following questions by using principles of molecular structure and intermolecular forces.

(a) Structures of the pyridine molecule and the benzene molecule are shown below. Pyridine is soluble in water, whereas benzene is not soluble in water. Account for the difference in solubility. You must discuss both of the substances in your answer.

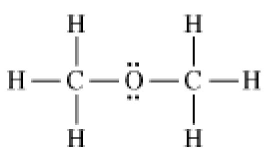


Pyridine

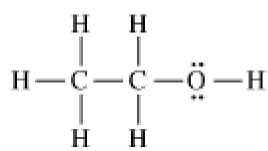


Benzene

(b) Structures of the dimethyl ether molecule and the ethanol molecule are shown below. The normal boiling point of dimethyl ether is 250 K, whereas the normal boiling point of ethanol is 351 K. Account for the difference in boiling points. You must discuss both of the substances in your answer.



Dimethyl Ether



Ethanol

(c) SO_2 melts at 201 K, whereas SiO_2 melts at 1,883 K. Account for the difference in melting points. You must discuss both of the substances in your answer.

(d) The normal boiling point of Cl_2 (l) (238 K) is higher than the normal boiling point of HCl(l) (188 K). Account for the difference in normal boiling points based on the types of intermolecular forces in the substances. You must discuss both of the substances in your answer.

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58. $$Mg(s) + 2\ H^{+}(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$$

A student performs an experiment to determine the volume of hydrogen gas produced when a given mass of magnesium reacts with excess HCl(*aq*) , as represented by the net ionic equation above. The student begins with a 0.0360 g sample of pure magnesium and a solution of 2.0 *M* HCl(*aq*).

a. Calculate the number of moles of magnesium in the 0.0360 g sample.

b. Calculate the number of moles of HCl(*aq*) needed to react completely with the sample of magnesium.

As the magnesium reacts, the hydrogen gas produced is collected by water displacement at 23.0°C. The pressure of the gas in the collection tube is measured to be 749 torr.

c. Given that the equilibrium vapor pressure of water is 21 torr at 23.0°C, calculate the pressure that the $H_2(g)$ produced in the reaction would have if it were dry.

d. Calculate the volume, in liters measured at the conditions in the laboratory, that the $H_2(g)$ produced in the reaction would have if it were dry.

e. The laboratory procedure specified that the concentration of the HCl solution be 2.0 *M*, but only 12.3 *M* HCl solution was available. Describe the steps for safely preparing 50.0 mL of 2.0 *M* HCl(*aq*) using 12.3 *M* HCl solution and materials selected from the list below. Show any necessary calculation(s).

10.0 mL graduated cylinder	Distilled water
250 mL beakers	Balance
50.00 mL volumetric flask	Dropper

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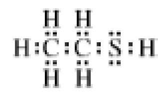
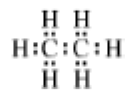
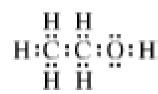
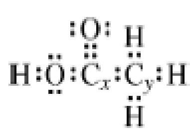
59. Use the information in the table below to respond to the statements and questions that follow. Your answers should be in terms of principles of molecular structure and intermolecular forces.

Compound	Formula	Lewis Electron-Dot Diagram
Ethanethiol	CH_3CH_2SH	
Ethane	CH_3CH_3	
Ethanol	CH_3CH_2OH	
Ethyne	C_2H_2	

a. Draw the complete Lewis electron-dot diagram for ethyne in the appropriate cell in the table above.

b. Which of the four molecules contains the shortest carbon-to-carbon bond? Explain.

c. A Lewis electron-dot diagram of a molecule of ethanoic acid is given below. The carbon atoms in the molecule are labeled x and y, respectively.



Identify the geometry of the arrangement of atoms bonded to each of the following.

i. Carbon *x*

ii. Carbon *y*

d. Energy is required to boil ethanol. Consider the statement “As ethanol boils, energy goes into breaking C−C bonds, C−H bonds, C−O bonds, and O−H bonds.” Is the statement true or false? Justify your answer.

e. Identify a compound from the table above that is nonpolar. Justify your answer.

f. Ethanol is completely soluble in water, whereas ethanethiol has limited solubility in water. Account for the difference in solubilities between the two compounds in terms of intermolecular forces.